

Calibration Procedures for Advance Analytik Sensors

1. VizSens pH (Analog) Sensor

Type: Manual 2-point or 3-point calibration using standard buffer solutions.

Steps:

1. Use fresh pH buffer solutions (pH 4.00, 7.00, and optionally 10.00).
2. Rinse the sensor with distilled water and gently blot dry.
3. Dip sensor into pH 7.00 solution. Allow reading to stabilize and set as midpoint calibration.
4. Repeat with pH 4.00 or 10.00 for slope calibration.
5. Adjust slope and offset via transmitter or handheld calibrator.
6. Rinse sensor and log calibration details.

2. VizSens ODO (Optical Dissolved Oxygen) Sensor

Type: Factory calibrated, optional 1-point field calibration.

Steps:

1. Ensure membrane cap is clean and intact. Allow stabilization.
2. Air Calibration: Expose to water-saturated air, stabilize, and set as 100%.
3. Zero-point: Use sodium sulfite solution, set as 0%.
4. Avoid frequent recalibration unless drift is observed.
5. Log calibration in maintenance records.

3. VizSens SS (Suspended Solids/Sludge) Sensor

Type: Zero-point and span calibration with grab sample comparison.

Steps:

1. Take grab sample and analyze SS in lab (e.g., gravimetric method).
2. Enter lab-determined SS value in transmitter.
3. Place sensor in clean water for zero-point if needed.
4. Clean sensor after calibration. Enable auto cleaning if applicable.

5. Document calibration values and actions.

4. Viz-Eco Duo / Viz-Eco Multi Transmitters

Calibration Interface:

1. Access via touchscreen or buttons: Sensor → Calibration.
2. Follow prompts to calibrate connected sensors (Analog/Digital).
3. Use known standard solutions or lab values for calibration.
4. Calibration logs stored in memory card (if installed).
5. Alerts for recalibration intervals can be enabled.

General Tips for All Sensors

- Calibrate regularly or monthly per SOP.
- Use traceable standard solutions.
- Clean sensors before and after calibration.
- Allow sensor to stabilize before recording values.
- Maintain calibration logs (date, user, value, method).